Cross-Network Intrusion Analysis

Phase 4 — Exploratory Data Analysis of the merged NetFlow data

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Notebook `07_eda_netflow.ipynb`

1 Data and method

Data. The merged NetFlow v3 table produced in phase 3 combines four networks: UNSW-NB15 (mixed lab network), CSE-CIC-IDS2018 (enterprise office network), ToN-IoT (IoT / industrial network) and BoT-IoT (IoT botnet network). It contains **17,551,848 rows standing for 63,012,224 flows**.

Unit of analysis: the behaviour. During cleaning, identical flows were merged into one row with a `count` column. Every statistic and test counts *behaviours* (rows): identical copies of a flow are not independent observations, and counting them would let the large floods dominate every result.

Research questions addressed.

- **SQ1:** which traffic characteristics distinguish normal traffic from attack traffic?
- **SQ2:** does the same attack family behave the same way in different networks?

Key variables. Four variables describe a flow: `IN_BYTES` (bytes sent by the client), `OUT_BYTES` (bytes sent back), `IN_PKTS` (packets sent) and `FLOW_DURATION_MILLISECONDS` (duration). The correlation analysis uses 12 traffic features.

Full data or sample. Counts, summary statistics, histograms and the Mann–Whitney test use *all* 17.5 million rows. Box plots, correlation heatmaps and the Kolmogorov–Smirnov test use *one* stratified sample: up to 10,000 random behaviours from each network × family group (187,418 rows, `random_state = 42`). Without it, Benign would fill two thirds of every plot and the smallest family (448 behaviours) would be invisible.

2 First look at the data

Table 1: Behaviours (rows) per family and network.

Family	UNSW	CIC-2018	ToN-IoT	BoT-IoT	All
Benign	1,307,351	8,178,232	2,282,642	21,951	11,790,176
Denial of service	5,360	642,600	573,277	642,187	1,863,424
Reconnaissance	13,134	–	225,074	898,728	1,136,936
Credential attacks	–	94,890	697,124	–	792,014
Web and injection	–	53	1,831,103	–	1,831,156
Malware and botnets	3,802	2,804	23,471	–	30,077
Exploitation and intrusion	42,740	27,213	–	–	69,953
Interception and spoofing	–	–	4,951	–	4,951
Fuzzing and crypto	32,713	–	–	–	32,713
Data theft	–	–	–	448	448
All behaviours	1,405,100	8,945,792	5,637,642	1,563,314	17,551,848
All flows	2,350,609	19,285,037	24,442,770	16,933,808	63,012,224

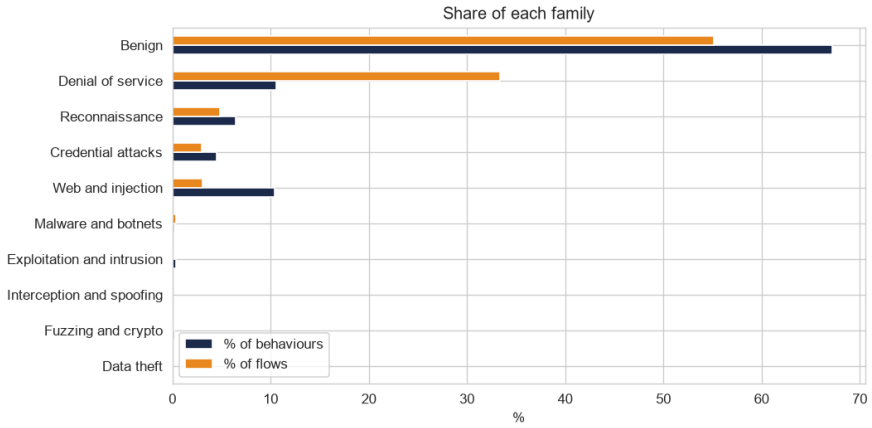


Figure 1: Share of each family in behaviours and in flows.

Interpretation.

- The data are complete: **no missing value**.
- **Benign is the majority** with 67% of the behaviours. Denial of service (11%) and Web and injection (10%) are the largest attack families; Data theft has only 448 behaviours.
- **Behaviours versus flows:** Denial of service is 11% of the behaviours but 33% of the flows, because a flood repeats the same flow many times. Counting behaviours prevents floods from dominating the analysis.
- **Not every family exists in every network.** Only Benign and Denial of service are present in all four. Fuzzing comes only from UNSW, Interception only from ToN-IoT, Data theft only from BoT-IoT, and Web and injection practically only from ToN-IoT. Comparisons across networks (SQ2) are therefore made on Denial of service.

3 Summary statistics

Table 2: Summary statistics of the key variables (all 17.5 million behaviours).

Variable	Mean	Median	Std	Q1	Q3	IQR	Skewness
IN_BYTES	3,725	701	142,790	164	1,623	1,459	357.2
OUT_BYTES	11,338	940	315,030	127	2,239	2,112	194.5
IN_PKTS	32.9	6	1,664	3	10	7	179.4
FLOW_DURATION_MILLISECONDS	14,549	755	33,205	47	4,057	4,010	2.4

Interpretation. All four variables are **strongly right-skewed**.

- The mean is much larger than the median (IN_BYTES: 3,725 against 701 bytes), because a few huge flows pull the mean up; the largest carries 189 MB.
- The skewness is far above 1 (357 for IN_BYTES), the limit above which a variable is "highly skewed".
- The standard deviation is larger than the mean.

The **median and the IQR** therefore describe a typical flow better than the mean and the standard deviation. A typical behaviour sends 701 bytes in 6 packets and lasts 0.75 s. The duration never exceeds 121 s, because the NetFlow exporter cuts longer connections into pieces.

Table 3: Median of the key variables per family (all behaviours).

Family	IN_BYTES	OUT_BYTES	IN_PKTS	Duration (ms)
Benign	1,135	1,572	8	1,325
Denial of service	528	1,136	6	10,812
Reconnaissance	44	40	1	23
Credential attacks	481	2,142	6	438
Web and injection	760	1,304	6	304
Malware and botnets	288	240	6	1,506
Exploitation and intrusion	822	555	10	800
Interception and spoofing	140	258	2	37
Fuzzing and crypto	812	336	10	708
Data theft	1,658	3,380	14	2,407

Interpretation. Each family has its own typical flow:

- **Reconnaissance** is the smallest (44 bytes, 1 packet, 23 ms): a single probe packet, typical of a scan.
- **Denial of service** lasts the longest (median 10.8 s): connections held open or flooded.
- **Data theft** moves the most data (1,658 bytes sent, 3,380 received, 14 packets).
- **Interception and spoofing** is short and small (140 bytes, 37 ms), like DNS messages.

The traffic variables therefore carry information about the type of attack (SQ1).

4 Plots of one variable

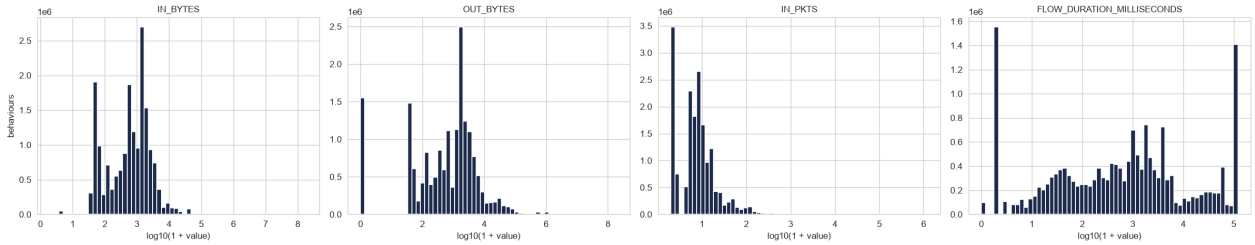


Figure 2: Histograms of the key variables, all behaviours. The axis shows $\log_{10}(1 + \text{value})$: each step of 1 means $\times 10$ ($2 = 100$, $3 = 1{,}000$), and the "+1" keeps the zeros on the plot.

Interpretation. None of the variables has a bell shape, even on a log scale. The distributions have several peaks and tall spikes at single values, each with a technical cause:

- `OUT_BYTES` $= 0$ for 8.8% of the behaviours: **no answer** (a scan of a closed port, a flooded victim);
- `IN_PKTS` $= 1$: one-packet flows (probes, DNS queries);
- `FLOW_DURATION_MILLISECONDS` piles up near 0 ms and at 120 s (the exporter's time limit).

These shapes already suggest that normality will be rejected (Section 7.1).

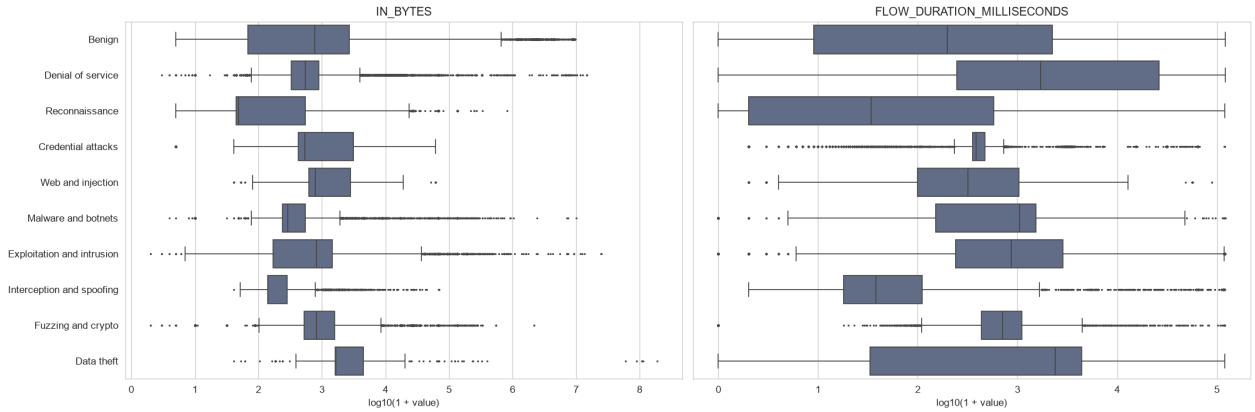


Figure 3: Box plots by family (stratified sample, log scale). Middle line = median, box = Q1 to Q3.

Interpretation. The families are clearly shifted from each other: Reconnaissance has the smallest byte counts and the shortest flows, Denial of service the longest flows, Data theft the most bytes. However, **the Benign box covers almost the whole range of the attack families**: no single variable separates normal traffic from all attacks, so a detection model will need several variables together. The points beyond the whiskers are mostly real attacks (large floods, long connections) and are kept.

5 Correlation analysis

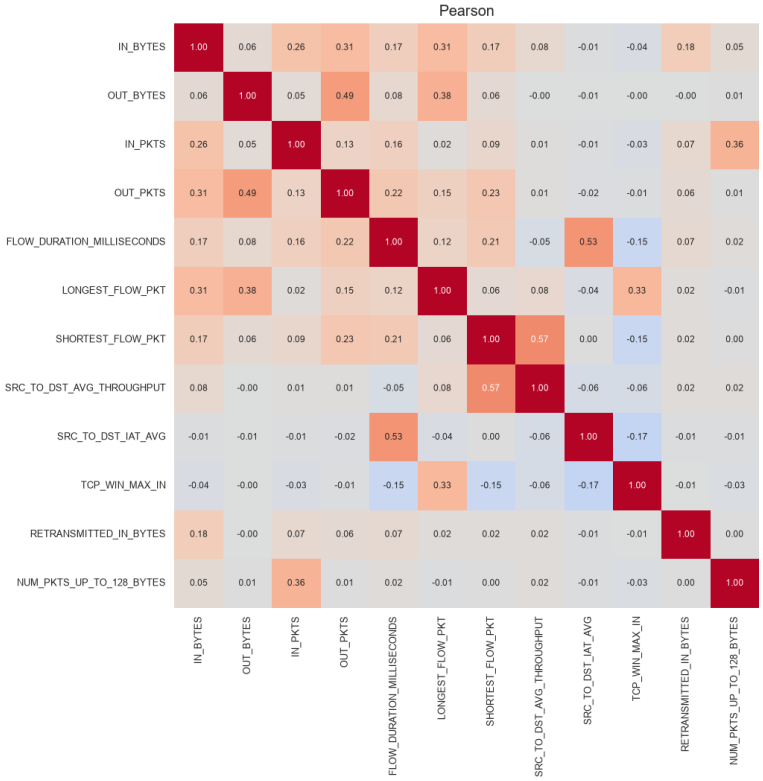


Figure 4: Pearson correlation of 12 traffic features (stratified sample).

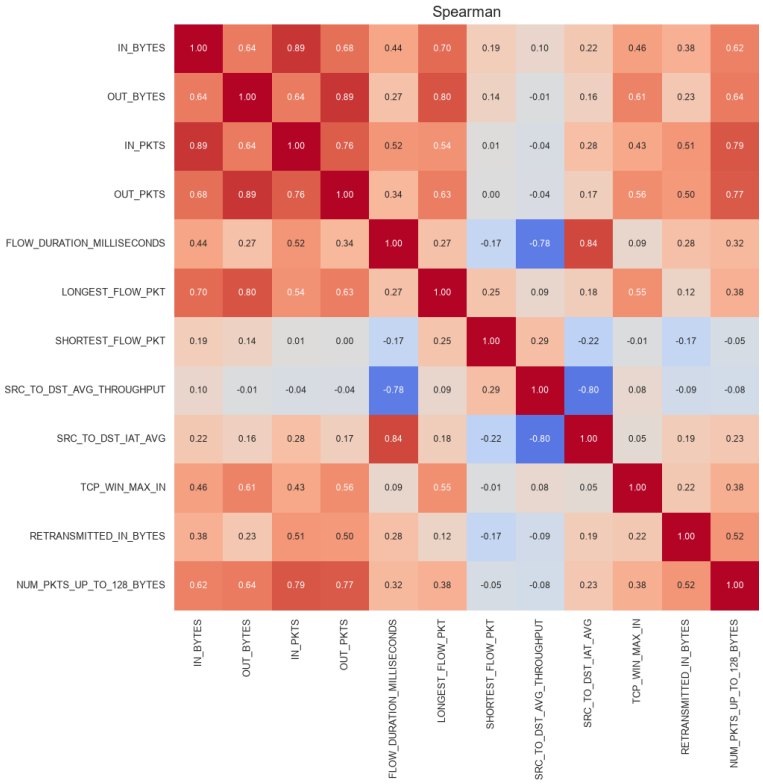


Figure 5: Spearman correlation of the same 12 features (stratified sample).

Table 4: Pairs where Pearson and Spearman disagree most, and the strongest Spearman pairs.

Variable 1	Variable 2	Pearson r	Spearman ρ
IN_BYTES	IN_PKTS	0.26	0.89
OUT_PKTS	NUM_PKTS_UP_TO_128_BYTES	0.01	0.77
SRC_TO_DST_AVG_THROUGHPUT	SRC_TO_DST_IAT_AVG	−0.06	−0.80
FLOW_DURATION_MILLISECONDS	SRC_TO_DST_AVG_THROUGHPUT	−0.05	−0.78
OUT_BYTES	OUT_PKTS	0.49	0.89
FLOW_DURATION_MILLISECONDS	SRC_TO_DST_IAT_AVG	0.53	0.84
OUT_BYTES	LONGEST_FLOW_PKT	0.38	0.80

Interpretation. The two heatmaps look very different. Pearson measures straight-line relationships and finds almost none, because a few gigantic flows dominate it. Spearman works on ranks, is not affected by extreme values, and finds strong relationships (for example IN_BYTES–IN_PKTS: $r = 0.26$ but $\rho = 0.89$). **Spearman is the right coefficient for this data.** The main relationships are:

- **volume:** bytes and packets rise together ($\rho = 0.89$ in both directions), and the more data a flow carries, the larger its biggest packet ($\rho = 0.80$);
- **time:** longer flows have longer gaps between packets ($\rho = 0.84$) and a lower throughput ($\rho = -0.78$).

Strongly correlated columns carry similar information; feature selection in phase 5 will keep one of each group.

6 Relationships between variables

6.1 Benign versus attack traffic in each network

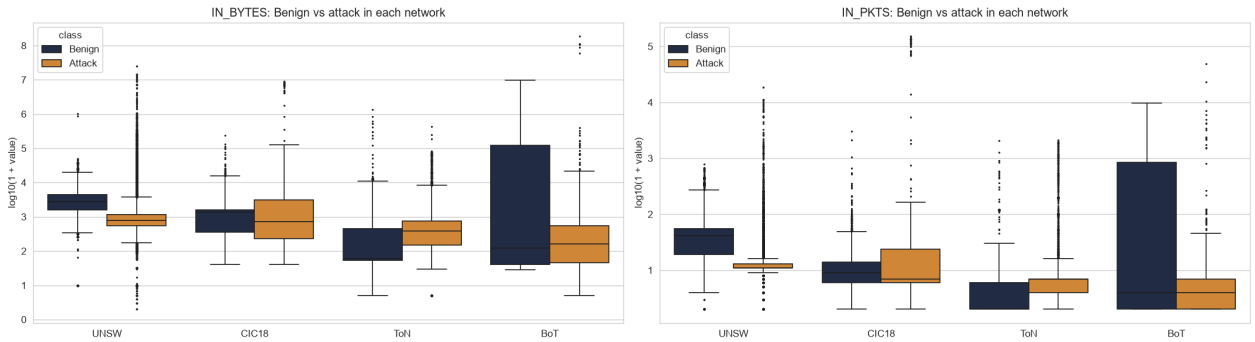


Figure 6: Bytes and packets sent, Benign against attack, in each network (stratified sample, log scale).

Table 5: Median of the key variables, Benign against attack, per network (all behaviours).

	IN_BYTES		OUT_BYTES		IN_PKTS		Duration (ms)	
Network	Benign	Attack	Benign	Attack	Benign	Attack	Benign	Attack
UNSW	2,750	814	8,824	336	40	10	95	716
CIC-2018	1,369	510	1,873	1,147	8	5	2,845	3,334
ToN-IoT	65	637	52	1,155	1	6	22	377
BoT-IoT	120	44	40	40	3	1	31	51

Interpretation. This is the most important relationship of the analysis: **the difference between Benign and attack traffic is not the same in every network; it even points in opposite directions.**

- **UNSW:** attacks are *smaller* than normal traffic (814 against 2,750 bytes, 10 against 40 packets).
- **ToN-IoT:** attacks are *larger* (637 against 65 bytes, 6 against 1 packet), because ToN-IoT's normal traffic consists mostly of tiny one-packet flows.
- **CIC-2018 and BoT-IoT:** the two classes are close.

Normal traffic itself differs strongly between networks (40 packets in UNSW, 1 in ToN-IoT). A rule such as "attacks send more bytes" holds in one network and fails in another. Test 1 (Section 7.2) checks this formally.

6.2 Destination port range and family

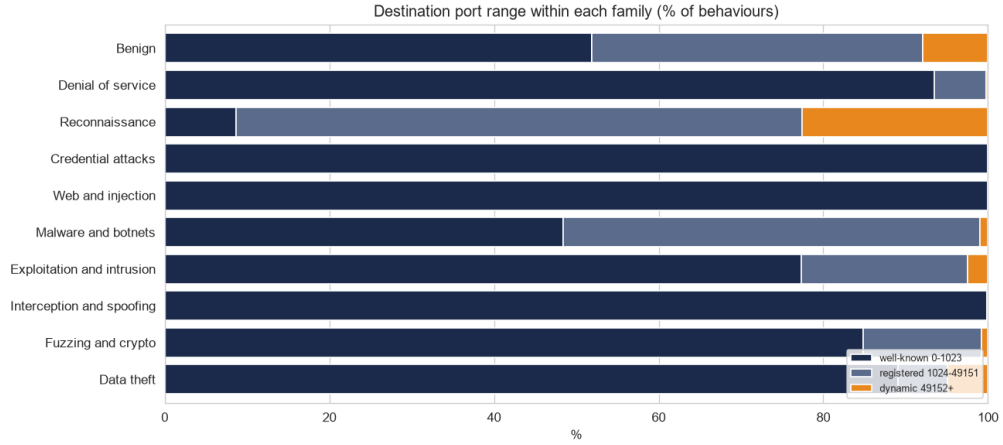


Figure 7: Destination port range within each family (% of behaviours, all data). Well-known 0–1023 = standard services (80 HTTP, 22 SSH, 53 DNS); registered 1024–49151; dynamic 49152–65535.

Table 6: Destination port range per family (row %).

Family	Well-known	Registered	Dynamic
Benign	51.9	40.1	8.0
Denial of service	93.5	6.3	0.2
Reconnaissance	8.7	68.8	22.5
Credential attacks	99.9	0.1	0.0
Web and injection	100.0	0.0	0.0
Malware and botnets	48.4	50.6	1.0
Exploitation and intrusion	77.4	20.2	2.4
Interception and spoofing	99.9	0.1	0.0
Fuzzing and crypto	84.8	14.5	0.7
Data theft	89.1	6.0	4.9

Interpretation. The port range is strongly linked to the family.

- Web, Credential, Interception and Denial-of-service traffic goes almost only to **well-known ports** (93–100%): these attacks target standard services.
- **Reconnaissance** goes 91% to registered and dynamic ports: a scan tries many ports, whereas a real service uses a few known ones.

- **Benign** traffic is mixed (52% well-known, 40% registered).

The destination port is therefore a useful variable for recognising the type of attack (SQ1); it will be encoded as a category in phase 5.

7 Hypothesis tests

7.1 Normality check: which kind of test can be used?

The t-test and ANOVA assume roughly normal data. Shapiro–Wilk and D'Agostino–Pearson test **H_0: the data come from a normal distribution**, on 5,000 random behaviours (the limit of Shapiro–Wilk), on the raw values and after $\log(1+x)$.

Table 7: Normality tests (5,000 random behaviours from the sample).

Variable	Scale	Skewness	Shapiro–Wilk p	D'Agostino p	Normal?
`IN_BYTES`	raw	18.95	2×10^{-94}	$< 10^{-300}$	no
	$\log(1+x)$	1.10	3×10^{-44}	3×10^{-238}	no
`OUT_BYTES`	raw	25.16	1×10^{-94}	$< 10^{-300}$	no
	$\log(1+x)$	−0.36	4×10^{-43}	9×10^{-37}	no
`IN_PKTS`	raw	48.85	2×10^{-95}	$< 10^{-300}$	no
	$\log(1+x)$	1.89	4×10^{-57}	$< 10^{-300}$	no
`FLOW_DURATION_MILLISECONDS`	raw	3.85	1×10^{-84}	$< 10^{-300}$	no
	$\log(1+x)$	−0.14	2×10^{-32}	2×10^{-22}	no

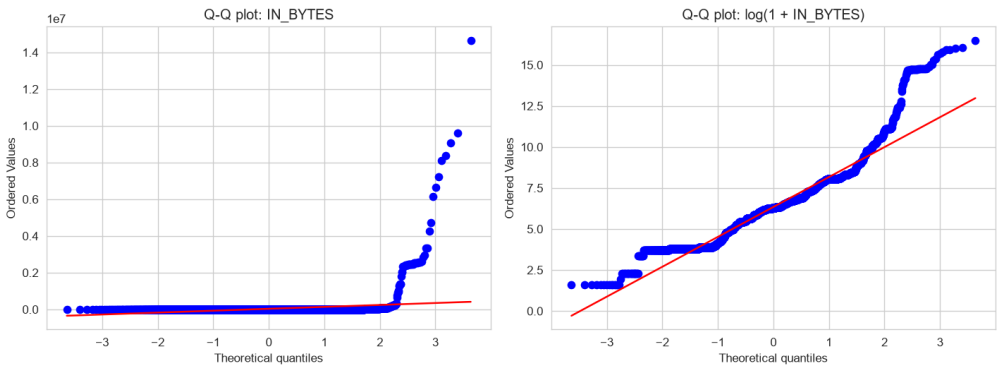


Figure 8: Q-Q plots of `IN_BYTES`, raw (left) and $\log(1+x)$ (right). Points on the line would mean a normal distribution.

Interpretation. Visually, no variable follows the normal curve. On the raw scale the Q-Q points lie flat along the bottom and then shoot up (a few huge values). After $\log(1+x)$ they come closer to the line but form a **staircase**, because many flows share exactly the same value. Both formal tests reject normality for every variable on both scales (all p-values below 10^{-20}). With 5,000 values the tests would also catch small, harmless departures, but here the departures are large: the raw skewness is between 4 and 49, and the plots show it clearly. The log transform helps the shape (`IN_BYTES` skewness 19 to 1.1) but does not make the data normal.

Practical conclusion: the t-test and ANOVA, which compare means and assume normal groups, are not suitable. The tests below use the **non-parametric** alternatives from the Chapter 4 slides, which work on ranks and need no normality. The CICIoT2023 analysis reached the same conclusion.

How the tests are read. With hundreds of thousands of rows, even a tiny difference gives $p \approx 0$. The p-value says *whether* there is a difference; the **effect size** says *how large* it is, and the conclusion

rests on it. Because several tests answer the same question, p-values are multiplied by the number of tests (**Bonferroni correction**) before being compared with $\alpha = 0.05$.

7.2 Test 1 (SQ1) — Mann–Whitney U: does attack traffic differ from Benign traffic?

Effect sizes used.

- Rank-biserial $r = \frac{2U}{n_1 n_2} - 1 \in [-1, 1]$: $r > 0$ means attacks are larger, $r < 0$ smaller. $|r| < 0.1$ negligible, < 0.3 small, < 0.5 medium, ≥ 0.5 large.
- In plain words, $P(\text{attack} > \text{Benign}) = (1 + r)/2$ is the chance that a random attack behaviour is larger than a random Benign behaviour (ties counted half); 50% means no difference.
- The 95% confidence interval of the difference in medians uses the distribution-free interval of each median (the sorted values at ranks $n/2 \pm 1.96\sqrt{n}/2$), which needs no normality.

Table 8: Mann–Whitney U: effect size r (attack against Benign) for the 16 tests.

Variable	UNSW	CIC-2018	ToN-IoT	BoT-IoT
`IN_BYTES`	−0.50	−0.11	+0.51	−0.10
`OUT_BYTES`	−0.73	+0.04	+0.69	−0.13
`IN_PKTS`	−0.62	−0.11	+0.59	−0.22
`FLOW_DURATION_MILLISECONDS`	+0.46	−0.05	+0.36	+0.12
Effect	large	negligible–small	large	small

Table 9: One example per network: medians, 95% confidence interval of the difference, and effect sizes.

Network	Variable	Median attack vs Benign (95% CI of diff.)	r	$P(\text{att.} > \text{Ben.})$	Effect
UNSW	`OUT_BYTES`	336 vs 8,824 bytes (−8,488 to −8,488)	−0.73	14%	large, smaller
ToN-IoT	`OUT_BYTES`	1,155 vs 52 bytes (+1,098 to +1,106)	+0.69	84%	large, larger
CIC-2018	`IN_BYTES`	510 vs 1,369 bytes (−861 to −858)	−0.11	45%	small
BoT-IoT	`IN_PKTS`	1 vs 3 packets (−2 to −2)	−0.22	39%	small

Interpretation.

- **H_0:** in each network, attack and Benign behaviours have the same distribution of the variable. **H_1:** the distributions differ.
- **Assumptions checked and test chosen (why):** the two groups are different behaviours, so they are independent. Normality is rejected (Section 7.1), so the two-sample t-test is not suitable and its rank-based alternative, the Mann–Whitney U test, is used. It runs on all behaviours of each network (from 21,951 to 8,178,232 per group).
- **Test and p-value:** 16 tests (4 networks × 4 variables); for example UNSW `OUT_BYTES`: $U = 1.75 \times 10^{10}$, $p < 10^{-300}$. All 16 Bonferroni-corrected p-values are below 10^{-180}.
- **Decision ($\alpha = 0.05$):** reject H_0 in all 16 tests.
- **Conclusion in plain language:** the size and the direction of the difference depend on the network.
  - In UNSW, a random attack flow receives more data than a random Benign flow only 14% of the time.
  - In ToN-IoT, it does so 84% of the time.

– In CIC-2018 and BoT-IoT, the effects are small.

The confidence intervals are very narrow because they use millions of behaviours: the medians are known almost exactly. Two cases show why the whole distribution matters, not only the median:

– in CIC-2018, `OUT_BYTES` has a lower median for attacks but r is slightly positive (+0.04), because the two distributions cross; the effect is negligible either way;

– in BoT-IoT, the confidence interval of the duration difference includes 0 (−88 to +32 ms): the medians may be equal, yet the test finds a small difference in the distributions ($r = +0.12$).

Answer to SQ1: attack traffic does differ from normal traffic, often strongly, but **which variable shows it and in which direction depends on the network**. A rule learned in one network ("attacks are small") can be wrong in another.

7.3 Test 2 (SQ2) — Kolmogorov–Smirnov: does the same family behave the same way in every network?

Family tested: Denial of service, the only attack family present in all four networks. **Effect size:** D = the largest gap between the two cumulative curves, from 0 (identical) to 1 (no overlap); $D < 0.1$ practically the same, 0.1–0.3 moderately different, > 0.3 clearly different.

Table 10: Kolmogorov–Smirnov on Denial of service (12 tests; all Bonferroni-corrected $p < 10^{-68}$).

	`IN_BYTES`		Duration	
Pair of networks	D	Medians (bytes)	D	Medians (ms)
UNSW vs CIC-2018	0.49	814 / 485	0.51	651 / 8,889
UNSW vs ToN-IoT	0.15	814 / 874	0.26	651 / 443
UNSW vs BoT-IoT	0.48	814 / 420	0.95	651 / 28,749
CIC-2018 vs ToN-IoT	0.48	485 / 874	0.59	8,889 / 443
CIC-2018 vs BoT-IoT	0.21	485 / 420	0.51	8,889 / 28,749
ToN-IoT vs BoT-IoT	0.54	874 / 420	**0.99**	443 / 28,749

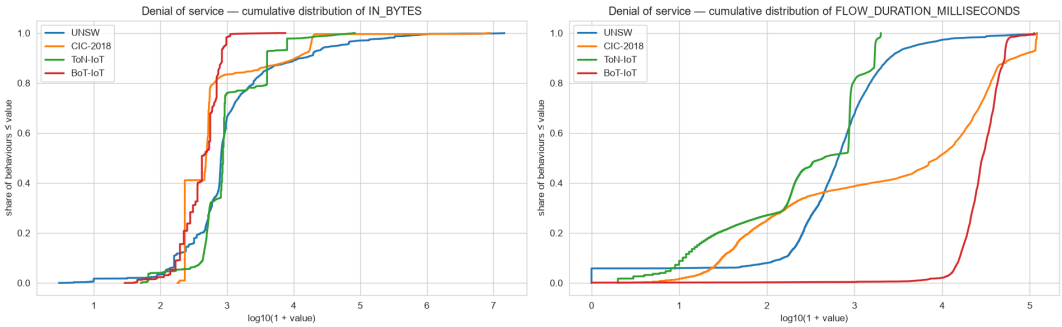


Figure 9: Denial of service: cumulative distribution of bytes and duration in each network. Identical behaviour would make the four curves lie on top of each other.

Interpretation.

- **H_0:** Denial-of-service flows have the same distribution in the two networks. **H_1:** the distributions differ.
- **Assumptions checked and test chosen (why):** the two samples come from different networks, so they are independent. Normality is rejected (Section 7.1). The question is

whether the whole behaviour is the same, not only the centre, so the two-sample Kolmogorov–Smirnov test is used: it compares the complete cumulative distributions. 6 pairs of networks $\times$ 2 variables = 12 tests, with up to 10,000 behaviours per network.

- **Test and p-value:** largest effect ToN-IoT vs BoT-IoT on the duration, $D = 0.99$, $p < 10^{-300}$; smallest effect UNSW vs ToN-IoT on the bytes, $D = 0.15$, $p = 3 \times 10^{-69}$. All 12 Bonferroni-corrected p-values are below 10^{-68}.
- **Decision ($\alpha = 0.05$):** reject H_0 in all 12 tests.
- **Conclusion in plain language:** a denial-of-service flow lasts a median of **0.4 s in ToN-IoT** but **29 s in BoT-IoT**; with $D = 0.99$ the two distributions barely overlap. Of the 12 comparisons, 9 are clearly different ($D > 0.3$), 3 moderately different and none the same. The cumulative curves of the four networks are far apart instead of lying on top of each other.

Answer to SQ2: the same attack family does **not** produce the same traffic in different networks. The testbeds used different tools, victims and settings, and these change the flows more than the attack type keeps them alike.

8 Wrap-up: what did the exploration tell us?

Three most important findings.

1. **SQ1:** attack traffic differs from normal traffic, but the direction depends on the network. In UNSW, attacks are smaller than Benign traffic (r down to -0.73); in ToN-IoT they are larger (r up to $+0.69$). No single variable separates attacks from normal traffic everywhere.
2. **SQ2:** the same attack family does not behave the same way in different networks. Denial of service lasts a median of 0.4 s in ToN-IoT and 29 s in BoT-IoT (KS $D = 0.99$), and none of the 12 comparisons finds the same distribution.
3. **The families have different profiles:** one-packet scans, long denial-of-service connections, large data-theft transfers. The destination port range is strongly linked to the family (scans use high ports, web and credential attacks use well-known ports). The variables do carry information about the type of attack.

Data-quality caveat. The traffic is far from normal (strong skewness, many repeated technical values such as the exporter's 120 s time limit). Some differences between networks therefore come from how each testbed was built and recorded, not only from the attacks. Because the data are so large, every p-value is close to 0, so the conclusions rest on the effect sizes (r, D).

Follow-up question. No single variable separates attacks from normal traffic in all four networks. Can a **combination** of variables do it, and how much accuracy is lost on a network the model has never seen (SQ3)?

Next steps (phase 5).

- Log-transform the volume and time variables.
- Encode the destination port as a category.
- Keep one variable from each strongly correlated group.